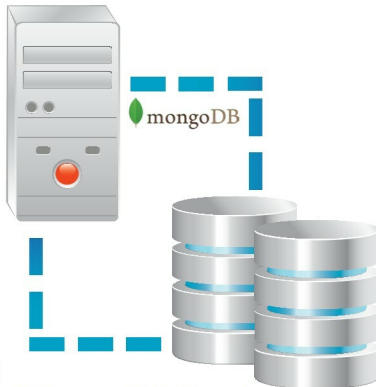




It's Easy to Scale Out a MongoDB Deployment

MongoDB is a unique NoSQL open source database, which is scalable and adaptable. It is an application of choice for many Fortune 500 companies and start-ups alike. In this, our second article in the series on MongoDB, discover how to scale out a MongoDB deployment.

MongoDB is a big name among NoSQL databases. One of its most significant features is that you can scale out your deployments quite easily, i.e., additional nodes can easily be added to the deployment to distribute data between them so that all data needn't be stored in one node. This concept is known as sharding, which when combined with replication, offers great protection against failover. So let's take a closer look at sharding and how it can be implemented.

I assume that you have MongoDB installed on your system. Well, just in case I am wrong, you can get it from <http://www.mongodb.org/downloads>. Download the zip file as per your operating system and extract the files once the download is completed. For this article, I am using MongoDB on a 32-bit Windows 7 system. Under the extracted directory, you will see a number of binary files. Among them, *mongod* is the server daemon and *mongo* is the client process.

Let's start the process by creating a lot of data directories that will be used to store data files. Now switch to the MongoDB directory that you extracted earlier. I have put that directory on my desktop (*Desktop\mongodb-win32-i386-2.6.0\bin*). So I'll open a command prompt and type:

```
cd Desktop\mongodb-win32-i386-2.6.0\bin.
```

Let's create the required directories, as follows:

```
mkdir .\data\shard_1 .\data\shard_2
mkdir .\data\shard_1\rs0 .\data\shard_1\rs1 .\data\shard_1\rs2
mkdir .\data\shard_2\rs0 .\data\shard_2\rs1 .\data\shard_2\rs2
```

Here we've created directories for two shards, with each having a replica set consisting of three nodes. Let's move on to creating the replica sets. Open another command prompt and type the following code:

```
mongod --port 38020 --dbpath .\data\shard_1\rs0 --replSet rs_1
--shardsvr
```

This creates the first node of the first replica set *rs_1*, which will be using port 38020 and *.\data\shard_1\rs0* as data directory. The *--shardsvr* option indicates that sharding will be implemented for this node. Since we'll be deploying all the nodes in a single system, let's just change the port number for additional nodes. Our first node is now ready and it's time to